El Equipo Interinstitucional de Ciencia y Aplicaciones (ISAT)

La Plata Taller

March 5-7, 2024

Land Surface Model - Water Balance - Rainfall-Runoff

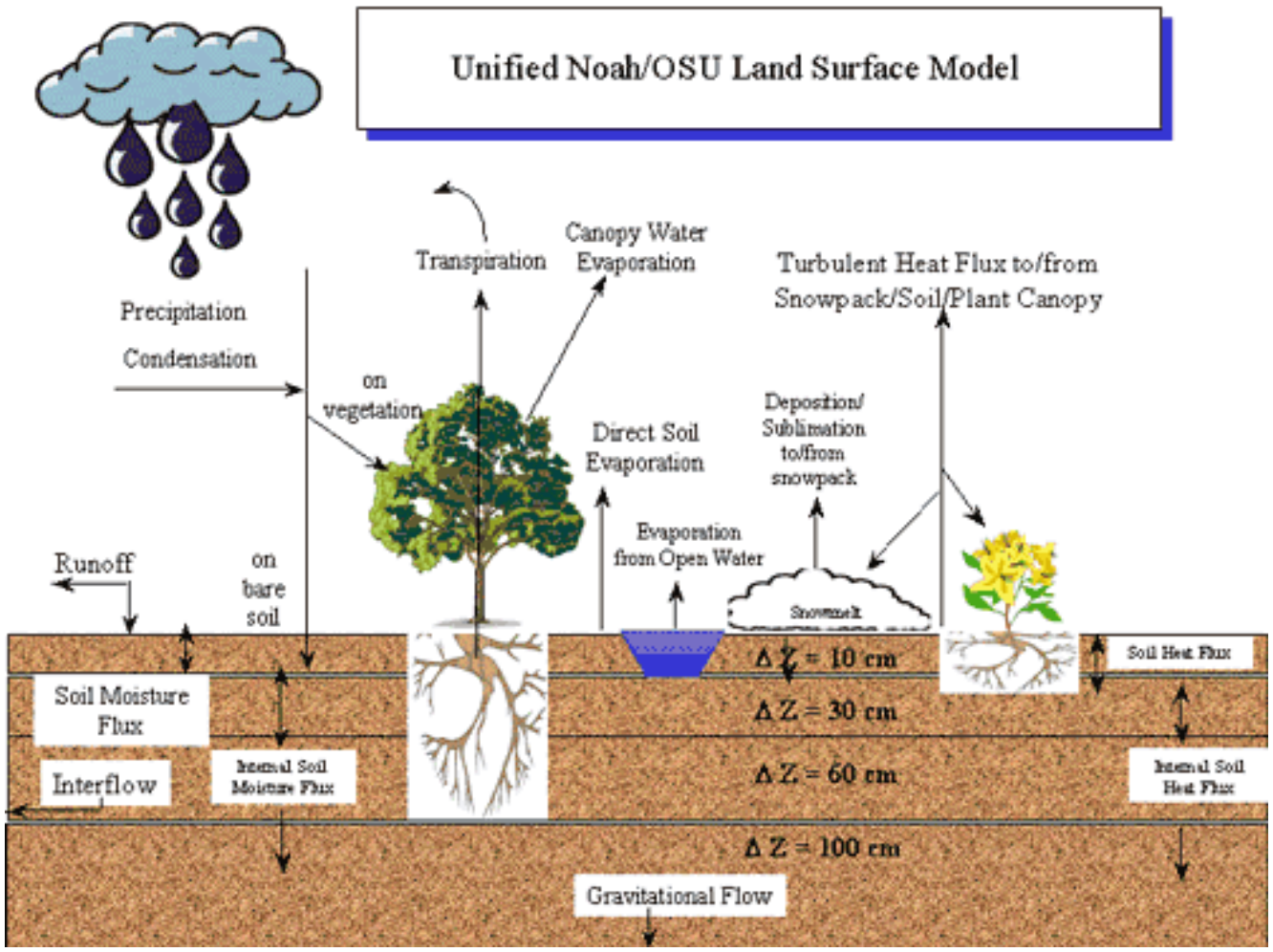
1. Water balance and rainfall-runoff methods typically used for hydrology/hydraulic models

2. Water Balance Equation: $P = Q + ET + \Delta S$
   - Where P = precipitation
   - Q = streamflow
   - ET = evapotranspiration
   - ΔS = change of water storage (soil moisture, snowpack, canopy water, groundwater, etc.)

3. Land Surface Models (LSM) used for coupled weather/climate models (account for land-atmosphere interactions)
   - LSMs compute both the mass and energy budget
   - More complex LSMs include dynamic vegetation, multi-layer snow physics, water table, groundwater, etc.

What is a Land Surface Model (LSM)?

Unified Noah LSM diagram (Chen and Dudia 2001)



- Land surface models analyze soil-water and soil-temperature relationships to estimate land-atmosphere energy exchange
- LSMs usually run gridded (a.k.a distributed) for large regions or for the entire globe
- Most current LSMs include a multi-layer surface configuration for estimating soil water/energy exchange (flux), multi-layer snow mass computations, dynamic vegetation models
- Complexity of the LSM may increase computing requirements

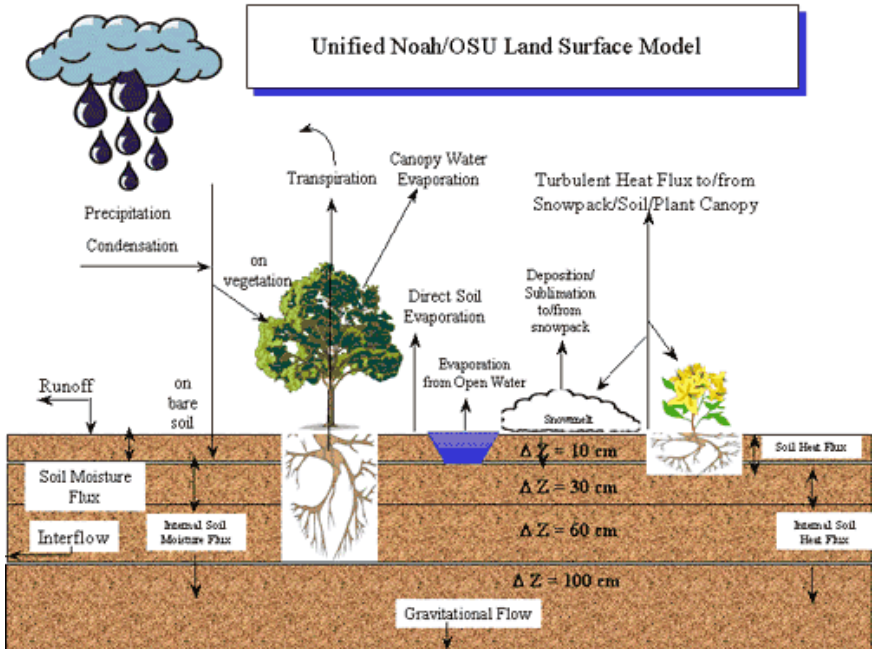
LSM Options in LIS

- The primary LSMs run in the Land Information System (LIS) include:
  - Noah – Versions 2.7.1, 3.X
  - Noah-MP (from National Center for Atmospheric Research, National Water Model) – Versions 3.6, 4.0.1
  - JULES (UK Met Office) – Versions 5.X
  - Catchment/CLSM (NASA Global Modeling & Assimilation Office) – Version F2.5
  - Variable Infiltration Capacity (VIC) (Univ. of Washington and Princeton Univ.) – Versions 4.1.1 and 4.1.2.l
  - Structure for Unifying Multiple Modeling Alternatives/Community Land Model (NCAR – in development)
- Any LSMs integrated into LIS are validated against a benchmark testcase
- LSMs added to LIS also include required software modifications needed to preprocess parameters datasets in the Land Data Toolkit (LIS preprocessing software)
- Multiple LSM options enable executing LIS-based ensemble modeling, allow us to represent uncertainties in the simulated results from different model physics and behavior
- LSM model configuration guide is posted at the end of the LIS online user's guide section 9.11: https://nasa-lis.github.io/LISF/LIS_users_guide/LIS_users_guide.html

Noah land surface model background

- Noah is a ”community” LSM, initially developed jointly by NOAAs **N**CEP, **O**regon State University, **A**ir Force Weather Agency, and NOAA Office of **H**ydrology (a.k.a. NOAH)
- Vertical soil water movement using Richard’s equation
- Infiltration is tied to precipitation rate (Mahrt and Pan, 1984) and soil texture
- Energy balance (sensible, latent fluxes) – Pan and Mahrt (1987)
- Thermal properties of ground impact water and energy flux rates

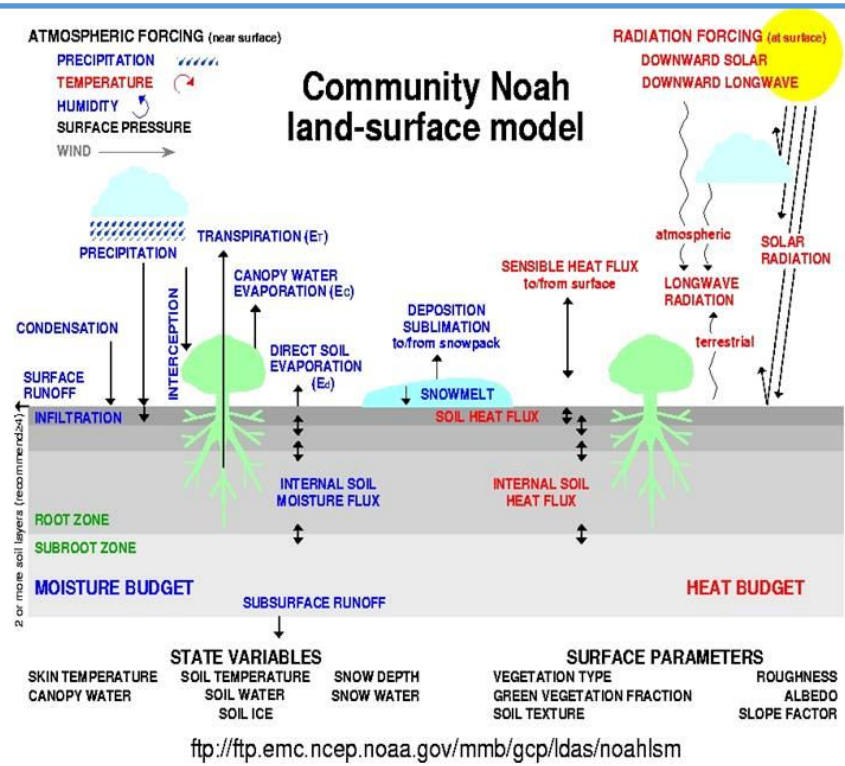
Unified Noah LSM diagram (Chen and Dudia 2001)



Noah-3.9 LSM

- Integrated the Noah LSM as in the WRF-3.9 software release
- There are two new physics options; we do not recommend using them at this time. By setting them as below, they will not be used, and will replicate the physics as in previous versions:

```
# The following lines are added for new options in Noah 3.9
Noah.3.9 thermal conductivity option:     1
Noah.3.9 FASDAS option (0 or 1):          0
```

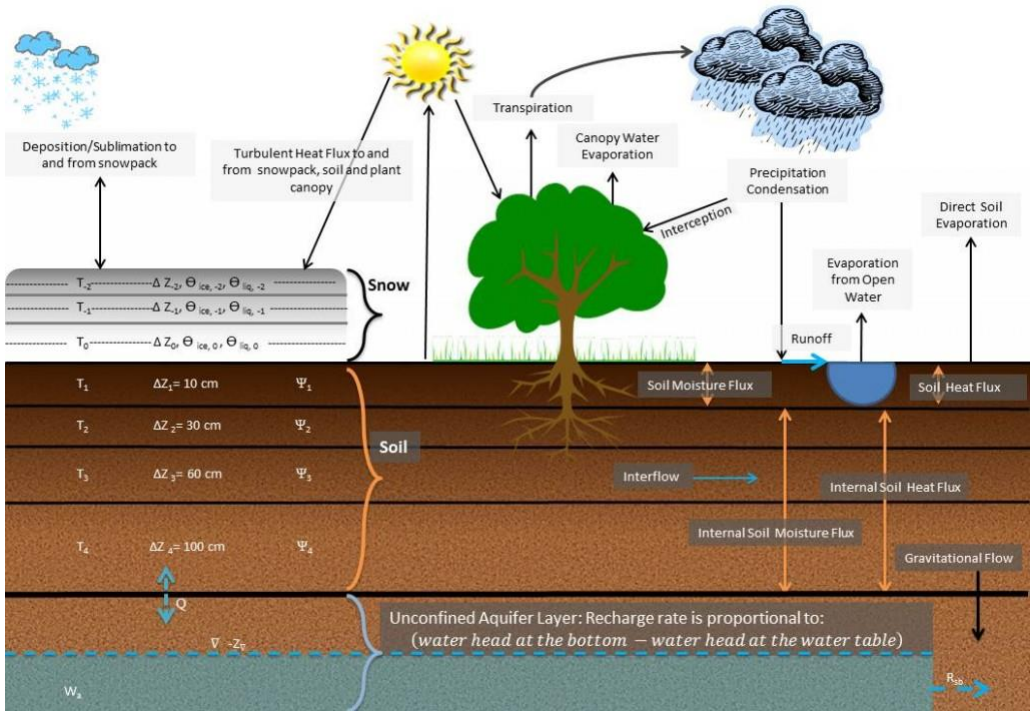


ftp://ftp.emc.ncep.noaa.gov/mmb/gcp/ldas/noahlsm

- Code contains several bug fixes from Noah-3.6:
  - Prevent discontinuous behavior for soil ice melting, and adds a dew term to QFX over snow cover.
  - Fixes high-frequency oscillations in latent heat flux seen in time series output over snow.
  - Fixed a minor bug in snow melting when the amount is small (snow cover fraction < 0.01).
  - Tweak to urban areas to maintain stability (making Noah behave more like Noah-MP LSM).
- Overall, only minor differences in code/simulation from Noah-3.6 to Noah-3.9

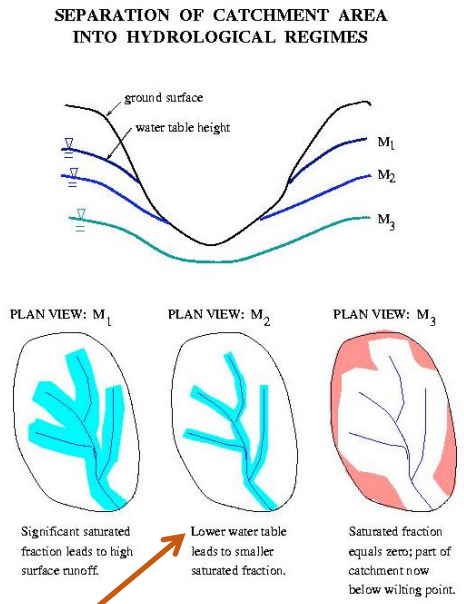
Noah-MP LSM

- Integrated the Noah-MP-4.0.1 LSM from the NCAR High Resolution LDAS system
- Also in WRF-4.0.1, and NOAA's National Water Model
- "MP" stands for Multi-Parameterization for several physics options, including options for prognostic phenology, groundwater, etc.
- Unlike Noah, Noah-MP uses:
  - 3-layer snowpack
  - Separate vegetation canopy
  - Advanced physics over glacier points

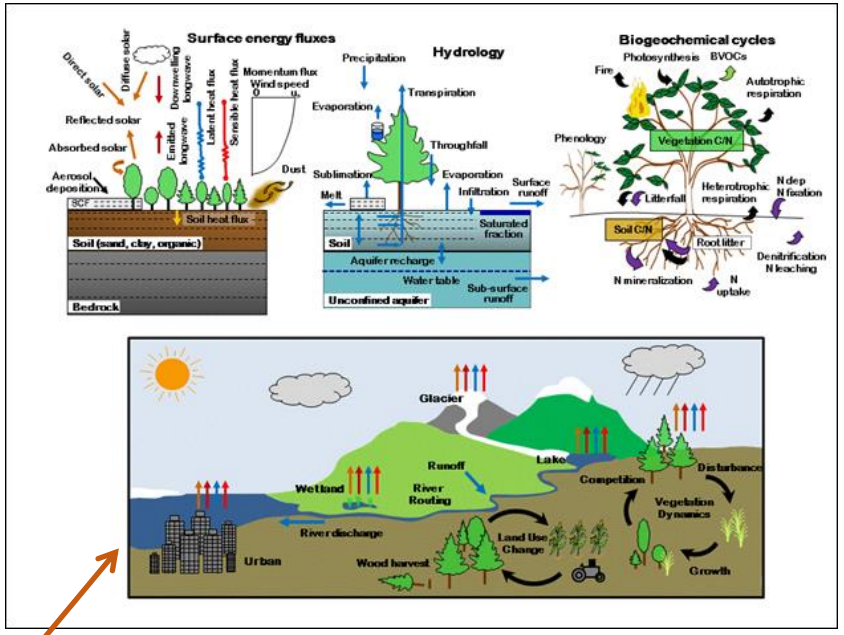


- Numerous physics configurations have been tested, and two main sets of physics options were evaluated – one using monthly climatological vegetation, and one using the prognostic phenology scheme.
- It is not possible to configure Noah-MP to run exactly as the Noah LSM.

NASA's Catchment LSM and NCAR CLM, SUMMA



Catchment LSM (CLSM) is developed by NASA/GMAO, and is the land-surface component of the NASA GEOS-5 Global Circulation Model.

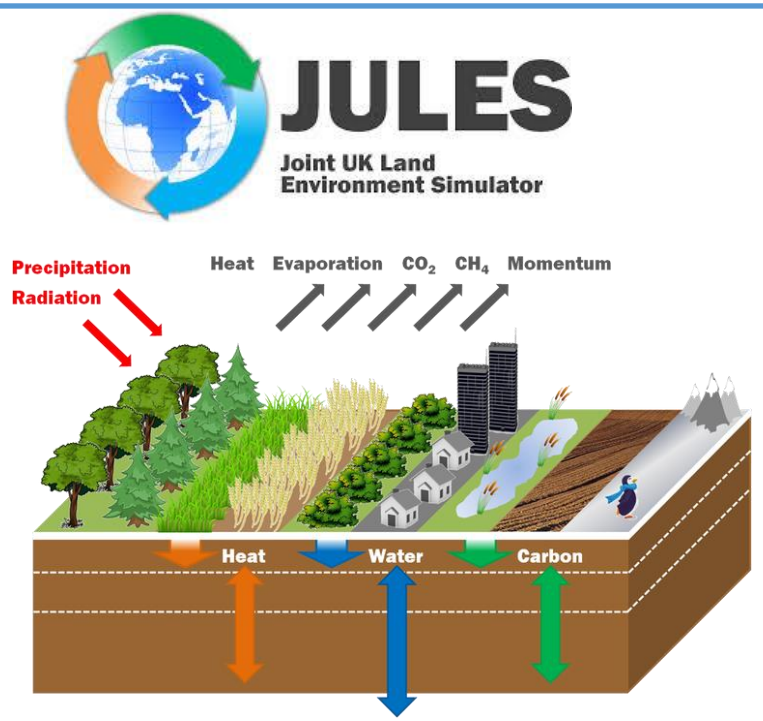


Community Land Model (CLM-4.5) is maintained by NCAR, and is the land-surface model for the Community Earth System Model (CESM).

VIC-4.1.2.l, SAC-HTET, and Noah-3.X are also in LIS and contain numerous upgrades.

JULES LSM

- Various versions of JULES (5.X) were integrated into LIS, and were tested with the 3-layer snow physics
- PS41 (Parallel Suite 41) includes GA8 and GL8 configurations for UM and JULES, respectively
- LIS-JULES PS41 configuration was based on the *rose-app.conf* file of GALWEM CONTINGENCY
- PS41 needs small changes due to differences in JULES between coupled and standalone mode
- JULES-5.0 (PS41) was run and evaluated in LIS in the global and regional domains

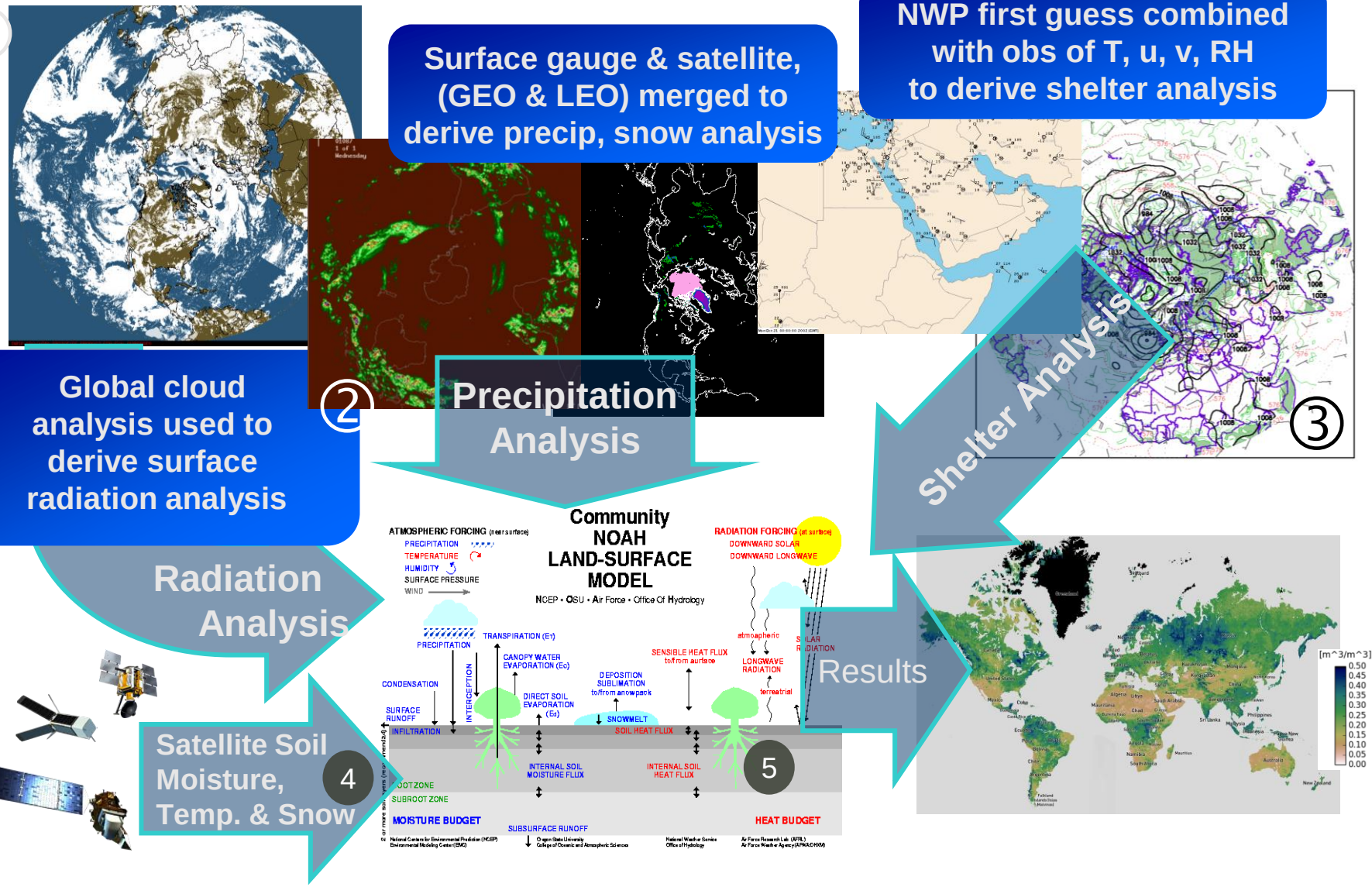


Switch	UM/JULES	LIS-JULES	Reason for change
l_albedo_obs	True	False	We don't have albedo data required to support in LIS-JULES standalone
l_embedded_snow	True	False	This option is not compatible with "aggregate" in standalone mode
l_snow_albedo	False	True	Need snow grain size for UM DUMP file with "embedded_snow" = False
Can_model	4	3	Canopy model 4 cannot be used with "aggregate" in standalone mode

Meteorology Forcings Overview

Land Data Assimilation System (LDAS) merges, blends, and assimilates data from multiple sources to compute soil-water mass and energy estimates used as input or “forcings” to the LSM

1. Global Cloud Cover Analysis
2. precipitation & snow observations & analyses
3. Weather (T, RH, Winds)
4. Satellite observations of soil moisture, temperature
5. Land surface model



Meteorological/Climate Forcings in LIS

- LIS philosophy: meteorological forcings (metforcing) ingested in its native format; LIS will interpolate (if needed) onto the compute domain
- LIS contains readers for products that contain the full suite of variables necessary to drive a LSM – (shortwave and downward longwave radiation, precipitation, temperature/humidity/wind/pressure) – as well as products with only 1 (or a few) variables
- LIS has an “overlay” feature, which allows users to combine different forcings. For example, one can use surface meteorology from NCEP Global Data Assimilation System, but “overlay” NASA IMERG gridded precipitation.
- Users can also “overlay” a global forcing dataset with a finer-resolution regional forcing

Forcing options in LIS

- “AGRMET”
  - USAF computed meteorological forcing
  - Raw data computing option - read in raw observations, satellite data, etc, compute gridded forcing fields (a.k.a “AGRMET Mode”)
- “Retrospective Mode”
  - Meteorological forcings computed by another agency, already available as a gridded product that can be read in by LIS
  - Global/regional: USAF, MERRA-2, ECMWF, GDAS, GFS, GEOS-5, NLDAS, GLDAS, Princeton, GSWP
  - Additional Precipitation Forcings that can be used
    - NASA TRMM
    - Integrated Multi-SatellitE Retrievals for GPM (IMERG)
    - Climate Hazards Group InfraRed Precipitation with Station Data (CHIRPS-2)
    - NOAA CMAP
    - CPC Morphing Technique (CMORPH)
    - Radar Products - Stage II/IV
  - Point observations(s): Bondville, IL, Loobos, PALS, USDA SCAN & SNOTEL network data

Forcing used for this workshop

- National Centers for Environmental Prediction (NCEP) Global Data Assimilation System (GDAS) gridded data
  - Computed by NCEP as part of their regular global model forecast process
  - The GDAS “03 HR forecast” and “06 HR forecast” files are also used by LIS to obtain accumulated precipitation and to interpolate between 3-hourly data down to the model timestep
- Gridded meteorological data is ~12km resolution global (T1534)